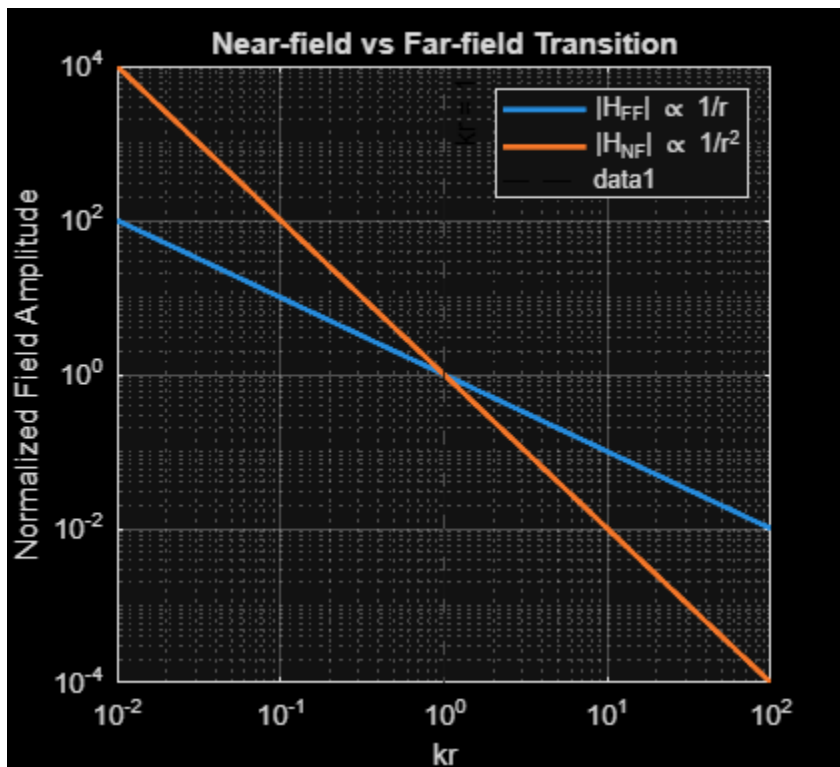

```
clc; clear; close all;
```

part a

```
kr = logspace(-2, 2, 1000); % from 0.01 to 100
H_FF = 1 ./ kr; % far-field
H_NF = 1 ./ (kr.^2); % near-field
figure;
loglog(kr, H_FF, 'LineWidth', 2); hold on;
loglog(kr, H_NF, 'LineWidth', 2);

xlabel('kr');
ylabel('Normalized Field Amplitude');
title('Near-field vs Far-field Transition');
legend('|H_{FF}| \proptopto 1/r', '|H_{NF}| \proptopto 1/r^2');
grid on;

xline(1, '--k', 'kr = 1');
```



part b

```
c = 3e8;
f = 1e9;
lambda = c/f;
r = lambda/(2*pi);
```

```
fprintf('Transition distance r = %.4f m\n', r);  
fprintf('In wavelengths: r = %.3f lambda\n', r/lambda);
```

```
Transition distance r = 0.0477 m  
In wavelengths: r = 0.159 lambda
```

Published with MATLAB® R2025b